

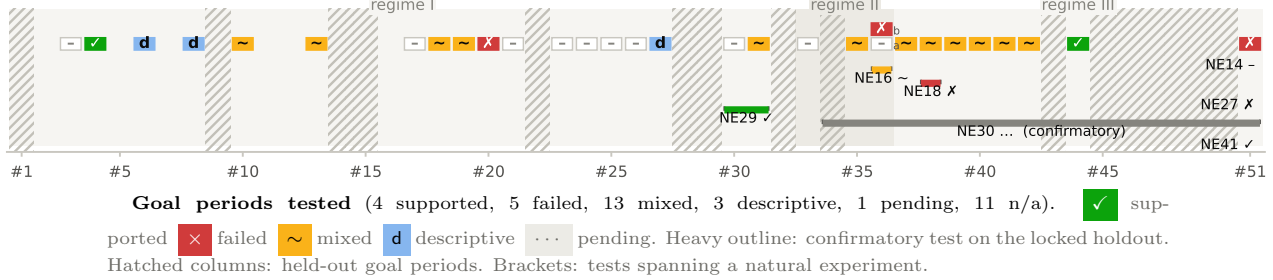
H15: Semantic information through natural scrambles: which information keeps agents and the swarm viable

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1.5/5 Usefulness 2.5/5

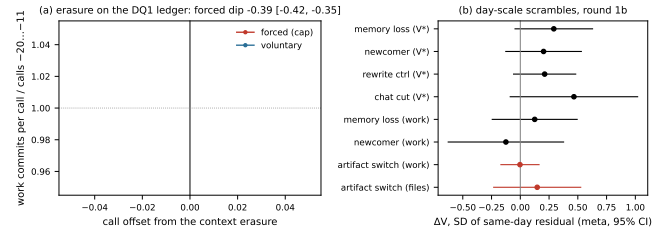
Bottom line. A forced wipe costs 39% of committed work for ten calls; the value is not in the few items read first (they carry 4% of the dip), and memory notes are about 11% enacted.

level: hypothesis · pipeline checks 85% · updated 2026-10-05



WHAT HAPPENED

The data is natural scrambles in non-reserved periods, per goal period, against placebo nulls calibrated on synthetic worlds built on the real village skeleton. **Rounds 1–1b:** losing 50–80% of memory (12 events) or starting with an empty memory (20 newcomers) shows no day-scale cost on the viability V^* (+0.29 SD $[-0.05, +0.63]$; newcomers +0.20). A forced context wipe (the 41-call cap) cuts work commits by 39% for ten calls (7/9 periods), whatever the agent saves to memory. **Round 2** asks where that value sits. It compares 20,592 forced wipes with 21,806 mid-segment pseudo-wipes in nine regime-III periods, and 37,031 consolidation notes with the commands that follow them.



Observables. Round 1b. (a) Work commits per call around a context erasure on the DQ1 ledger, relative to calls $-20 \dots -11$. A forced wipe (41-call cap) cuts committed work by about 39% for ten calls. Voluntary consolidations dip more after a work burst (task phase). (b) Day-scale scrambles on V^* and on work output (meta over goal periods, 95% CI). Only the chat cut (3 events) departs from zero.

PHYSICAL MODEL

Model 04, Kolchinsky–Wolpert semantic information. The system X is one agent: its memory M (day-scale store) and its context window C (session-scale store, filled by chat and events). Y is the rest of the village. A scramble erases M or C , or cuts $Y \rightarrow X$; $\hat{p}$ is the scrambled joint. V is viability per agent-day (5 candidates) or per call (work). u is its same-day residual, and δ is the fraction removed. S is the semantic information and η its share of $I(X; Y)$.

$$\Delta V = V[p] - V[\hat{p}], \quad \kappa = \Delta V / \delta$$

$$S = \min_f I(\hat{p}^f) \text{ s.t. } V[\hat{p}^f] = V[p], \quad \eta = \frac{S}{I} \leq 1 - \delta^*$$

$$u_{ad} = V_{ad} - \bar{V}_{-a,d}$$

$$\widehat{\Delta V} = \bar{u}_{\text{post}} - (\mu_a + \rho^s \mathbb{E}[h_{t_0-1} | u_{\text{pre}}])$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The working context keeps an agent productive, and its value is spread over the working state, not stored in a few items. Re-showing the last files or the newest messages after a wipe recovers at most a few percent of the lost work (4% $[-23, +26]$). Agents re-read by habit, not by what was erased. The usable number stays the cost of a reset: -39% of committed work for about ten calls, about a tenth of output at a 41-call cadence. To recover it, raise the cap; a carried summary note will not do it, because notes are only about 11% enacted and their content buys no output. The first-read share $\eta_{\text{first}} \approx 0.04$ is the closest this data comes to Kolchinsky–Wolpert’s η .

RESULTS VS PREDICTIONS

Prediction	Result (non-reserved)	Verdict
P1 memory loss $\leq +0.29$ $[-0.05, 0.63]$		×
−0.25 (1b)		
P4 newcomers pay $+0.20$; progress $+0.31$ $[0.06, 0.56]$		×
(1b)		
P5' forced-wipe dip work -0.39 $[-0.42, -0.35]$ (7/9)		✓
(1b)		
R1a re-read $\geq +0.05$	$+0.049$ $[0.036, 0.063]$; re-touch mixed 0.33 vs 0.32	
R1b trail softens dip	T2 1.19 $[0.99, 1.48]$; T1 1.26; 7-d inconcl. 1.03	
R1c memory-loss re-read	$+0.04$ $[-0.05, 0.13]$, $n = 32$	inconcl.
R2a last files ≥ 1.15	1.10 $[1.01, 1.21]$	partly
R2b new messages ≥ 1.14 $[1.02, 1.28]$; DL 1.02 1.10		mixed
R2c note names repo ≈ 1	1.04 $[0.92, 1.15]$	✓
R2d first reads $\geq 25\%$ of dip	0.04 $[-0.23, 0.26]$	×
R3a notes performative	excess 0.090 $[0.076, 0.104]$; novel 0.043	×
R3b notes retrospective	pre-post $+0.010$ $[0.004, 0.016]$	✓

GOAL PERIODS IN DETAIL

NE30	...	pending (confirmatory; not run) (holdout)
G04	✓	1b: supported (V* V_eng; MN -0.50 (z -0.7); 2026-10-0...
G06	d	1b: descriptive (V* V_eng; MR +0.40 (z +0.5); 2026-10...
G08	d	1b: descriptive (V* V_eng; MR -0.69 (z -1.0); 2026-10...
G10	~	1b: mixed (V* V_eng; MN +0.49 (z +1.5); 2026-10-04, r...
G13	~	1b: mixed (V* V_eng; ML +0.40 (z +1.2); 2026-10-04, r...
G18	~	1b: mixed (V* V_eng; MN +0.58 (z +1.2); 2026-10-04, r...
G19	~	1b: mixed (V* V_eng; MN +0.96 (z +1.2); 2026-10-04, r...
G20	×	1b: failed (V* V_eng; MN +0.56 (z +1.1); 2026-10-04, ...
G27	d	1b: descriptive (V* V_eng; MR +0.10 (z +0.2); 2026-10...

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

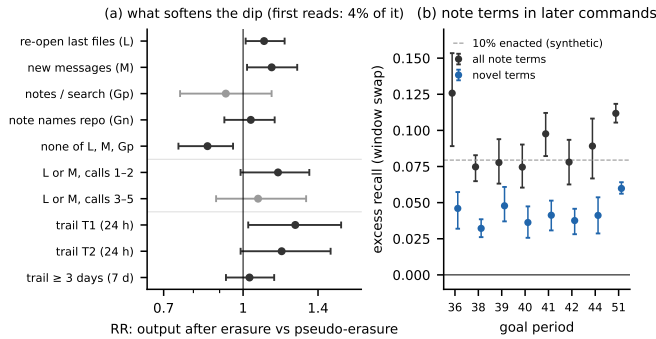
Day-scale and memory-loss tests see only large effects (power 17–41%; shifts ≥ 0.1). The erasure dip is post hoc in origin; NE41 replicated it with dated predictions, but the reserved-data check (C3) has not run. The round-2 classes are chosen by the agent; the pseudo-wipe arm removes their association with output when nothing is erased, under a parallel-trends assumption. New messages carry value only in the pooled fit (G51-driven; across periods $\tau = 0.17$). Notes are the next-session goal, not the full memory file; recall is a term-overlap proxy.

ROUND 2

Re-freeze `confirm_ne30.py` on work commits and run C3.

- **Essence**. Where does the swarm keep its knowledge? In artifacts and history, not in agents.
- **H15-R1** Stigmergic memory: after a memory loss, agents re-acquire state from artifacts (file reads, history search) within a few turns; richer artifact trails mean faster recovery (E3).
- **H15-R2** Semantic information is concentrated in the few items read first after a reset (the goal and the last messages).
- **H15-R3** Memory notes are performative: their content rarely reappears in later actions.

H15-semantic-information-scrambles · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H15-semantic-information-scrambles/README.md · ratings: H15 round-2 agent (suggested); round 1 calibrated by coordinator 2026-10-04 (page 2)



Second figure. Round 2. (a) Ratio of ratios: how much more each class raises work output after a forced wipe than after a pseudo-wipe (pooled fit, agent-day bootstrap 95% CI; gray: unpowered). The first two calls' reads (last files, new messages) add 10–14% but remove only 4% of the dip. A richer repo trail shows no dose order. (b) Excess recall of consolidation-note terms in the next segment's commands over the same agent's windows ≥ 3 days away, per period. Dashed: the synthetic level of 10% of a note enacted.